Assignment 1 Pseudocode

Q1. Define a function "solution" with a list input parameter 'A', and an integer 'K' (number of rotations):
  Run a for loop that goes from 0 to K:
    Define a new variable "value"
    Pop last stack 'A' and hold the value of that index in A
    Convert A into a double-ended queue
    Append "value" to the beginning of 'A'
    Convert A back into a list
  Return 'A' after the loops completion

Q3. Define a function "solution" with input parameters 'X', 'Y' and 'D' (starting position, ending position, distance increased each step):
  Define a variable to measure the number of loops that have passed (num_loops)
  Run a while loop that continues as long as X < Y:
    Increase X by D
    Increase num_loops by 1
  Return num_loops.

Q8. Define a function "solution" with a list input parameter 'A':
  Define a variable that measures the number of cars that have passed (num_passing)
  Run a for loop with variable x that starts at zero and ends at the length of 'A':
    Check if the value of A at index x is 1 (going west) and if true, continue
    Create a for loop with a variable y that starts x+1 and ends at the length of 'A':
      If the value of A at y isn't equal to the value of A at x (going opposite directions):
        increase num_passing by one
  Return num_passing

Q9. Define a function "insertion_sort" with a list parameter 'A':
  Run a for loop with variable x that ends at the length of 'A' - 1:
    Run a for loop with variable y that starts at the value of x, ends at -1 and decreases by 1 each loop:
      If the value of 'A' at y+1 is less than the value of A at y:
        Create a temporary variable to store the value needed to be swapped
        Set "temp" equal to the value of 'A' at y+1
        Set the list 'A' at index y+1 equal to the value of 'A' at y
        Set the list 'A' at index y to the value stored in the temporary variable
      Else:
        Break the loop (y)

  Define a function "solution" with a list parameter 'A':
    Call insertion_sort with the input parameter 'A'
    Run a for loop with variable p that starts at zero and ends at the length of 'A' - 1:
      Run a for loop with a variable q that starts at p+1 and ends at the length of 'A' - 1:
        If the value of 'A' at index q is less than or equal to the value of 'A' + 1 at index q, break the loop (q)
        Else, return the value of 'A' + 1 at index p

Q10. Define a function "make_zero_list" with an integer parameter 'N' (length of the list):
  Define an empty list variable
  Run a for loop that starts at zero and ends at 'N'
    Append a zero to the list
  Return the list

  Define a function "solution" with an integer parameter 'N' and a list parameter 'A' (length of the zero list):
    Call "make_zero_list" with an input variable 'N'
    Create a variable 'B' and set it equal to the list created in "make_zero_list"
    Run a for loop with a variable 'K' that starts at zero and ends at the length of the list 'A':
      If the value of 'A' at index K is less than or equal to 'N':
        Increase the value of 'B' at the index of the value of 'A' at index K (-1 so as to correct the boundaries) by one
      Else:
        Create a temporary variable to determine the maximum value in the zero-list (max) and set it equal to zero (the default value)
        Run a for loop with a variable "scan" that starts at zero and ends at the length of list 'B':
          If the value of 'B' at index "scan" is greater than the current value of "max":
            Set max equal to the value of 'B' at "scan"
        Run a for loop with the variable "assign" that starts at zero and ends at the length of 'B':
          Set the value of 'B' at index "assign" to max
    Return 'B'